

CUSTOMER SUPPORT TICKET PRIORITY PREDICTION AND AUTOMATED ASSIGNMENT SYSTEM USING AGENTFORCE

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Team Size : 5

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User Story

The Customer Support Ticket Priority Prediction and Automated Assignment System uses Salesforce and Agentforce to analyze customer support ticket descriptions, automatically determine priority as High, Medium, or Low, and trigger backend automation for appropriate support handling. The solution uses the Support Ticket Intelligence custom object, an Auto-Launched Flow, and an Agentforce subagent named Support Ticket Priority Analysis. High-priority tickets create an urgent handling task and are assigned to a senior support agent.

1. INTRODUCTION

1.1 Project Overview

Customer support teams receive a large number of tickets daily, while prioritization and assignment can be manual. This project provides a Salesforce-based solution that analyzes ticket information, classifies urgency, assigns the appropriate support level, and initiates automation for critical cases.

1.2 Purpose

- Automatically classify support tickets as High, Medium, or Low priority.
- Retrieve the latest support ticket associated with a customer Account.
- Reduce manual prioritization and assignment effort.
- Create an urgent handling task for High-priority tickets.
- Use Agentforce to analyze ticket details and trigger backend automation.
- Improve ticket response and resolution efficiency.

2. IDEATION PHASE

2.1 Problem Statement

PS	User	I'm trying to	But	Which makes me feel
PS-1	Support Agent	focus on critical tickets first	tickets are not always prioritized automatically; prioritization can be manual	frustrated
PS-2	Customer	get urgent issues resolved quickly	critical issues may wait behind lower-priority tickets	concerned
PS-3	Support Manager	assign tickets to the right support level	assignment may require manual decisions	overwhelmed
PS-4	Support Agent	identify urgent issues from descriptions	important keywords can be missed	uncertain
PS-5	Support Manager	reduce SLA-related risk	older unresolved tickets may remain unnoticed	concerned

2.2 Empathy Map Canvas

User Persona: Support Agent

- Think & Feel: Wants critical issues identified quickly so urgent customer problems receive attention.
- Hear: Receives customer issue descriptions and support requests.
- See: Ticket status, issue type, description, priority level, assignment, and SLA risk.
- Say & Do: Reviews tickets, handles assigned cases, and focuses on urgent support issues.
- Pain: Manual prioritization, missed urgency signals, and delays in handling critical tickets.
- Gain: Automatic priority classification, clearer assignment, urgent-task creation, and reduced manual effort.

2.3 Brainstorming & Idea Prioritization

- Manual ticket registers
- Spreadsheet-based ticket tracking
- Standalone support portal
- Salesforce CRM with Agentforce

Final Selection: Salesforce CRM selected due to automation through Salesforce Flow, AI-assisted ticket analysis through Agentforce, automatic priority classification, automatic urgent-task creation, and reduced manual support effort.

3. REQUIREMENT ANALYSIS PHASE

3.1 Customer Journey Map

Ticket Creation → Account Identification → Ticket Retrieval → Description Analysis → Priority Classification → Agent Assignment → Urgent Task Creation → Support Handling → Resolution

3.2 Solution Requirements

Functional Requirements

FR No.	Requirement
FR-1	Support ticket creation and management
FR-2	Account and Contact association
FR-3	Ticket description analysis
FR-4	Automatic priority classification
FR-5	High-priority task creation
FR-6	Support agent assignment
FR-7	SLA breach risk checking
FR-8	Agentforce conversational analysis

Non-Functional Requirements

NFR	Description
Usability	Clear Agentforce interaction and understandable priority messages
Reliability	Consistent priority classification based on configured conditions
Performance	Quick retrieval and analysis of the latest support ticket
Security	Controlled Salesforce access to customer and ticket information

Maintainability	Flow and Agentforce configuration can be updated
Scalability	Supports increasing support-ticket volumes through Salesforce automation

3.3 Data Flow Diagram

Customer / Support Request → Account → Latest Support Ticket → Ticket Description → Analyze Description → Priority Level → High/Medium/Low Decision → Agent Assignment → High-Priority Task (if applicable) → Final Response

Agentforce: User provides Account Name → Agentforce Support Ticket Priority Analysis → Auto-Launched Flow → Ticket Analysis → Priority and Assignment Output → Response to User.

3.4 Technology Stack

Layer	Technology
Platform	Salesforce
Database	Support Ticket Intelligence __c with Account, Contact, User and Task records
Logic	Auto-Launched Flow
AI	Agentforce
Agent Configuration	Support Ticket Priority Analysis subagent
Automation	Flow Decision, Assignment and Create Records
Output	Agentforce conversation response

4. PROJECT DESIGN PHASE

4.1 Problem–Solution Fit

Problem	Solution	Benefit
Manual ticket prioritization	Flow analyzes description and sets priority	Faster identification of urgent cases
Urgent issues may be missed	High-priority keyword decision	Critical cases receive attention
Manual task creation	Flow creates Task for High priority	Reduced manual effort
Manual support assignment	Flow sets assigned support level	More consistent handling
SLA risk can be overlooked	Optional SLA Risk decision	Better risk awareness
Need for conversational access	Agentforce subagent AI-assisted interaction	Conversational support handling

4.2 Proposed Solution

Parameter	Description
Problem	Manual ticket prioritization and assignment causing delays in handling critical customer support issues
Solution	Salesforce-based Customer Support Ticket Priority Prediction and Automated Assignment System
Innovation	Agentforce AI + Auto-Launched Flow for analyzing ticket descriptions, classifying priority, assigning support level, and triggering urgent-task automation
Impact	Faster identification of critical tickets, reduced manual effort, improved response time, and better support-team productivity
Business Model	Customer support operations involving customers, support agents, support managers, and AI-assisted ticket handling
Scalability	Supports increasing support-ticket volumes through

	Salesforce automation, with scope for additional assignment rules, SLA monitoring, and priority indicators
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4.3 Solution Architecture

Users / Support Team → Agentforce → Support Ticket Priority Analysis → Auto-Launched Flow → Account & Ticket Data → Description Analysis → Priority Decision → Task / Assignment → Final Message → Agentforce Response

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

Agile methodology is used with Epics, User Stories, Story Points, and sprint-based execution. Velocity calculation is used for estimation.

Product Backlog

Sprint / Epic	User Story	Points	Priority
Sprint-1 Developer Setup	USN-1: Configure Salesforce environment for Support Ticket Intelligence automation solution	3	High
Sprint-2 Data Modeling	USN-2: Create Support Ticket Intelligence object with required fields and relationships	5	High
Sprint-2 Data Modeling	USN-3: Associate support tickets with Accounts and Contacts	3	High
Sprint-3 Automation	USN-4: Create Auto-Launched Flow to retrieve latest ticket and analyze description	5	High
Sprint-3 Automation	USN-5: Classify tickets as High, Medium, or Low using urgency conditions	5	High
Sprint-4 Agentforce & Automation	USN-6: Analyze support ticket details and trigger backend Flow using Agentforce	8	High
Sprint-4 Task & Assignment	USN-7: Create urgent task and assign appropriate support level for High priority	5	High
Sprint-5 SLA Management	USN-8: Check SLA breach risk for older unresolved tickets	5	Medium
Sprint-6 Agentforce & Conversation	USN-9: Provide Account Name and receive ticket ID, priority, assigned agent, and action message	5	Medium

6. PROJECT DEVELOPMENT PHASE

Milestone 1: Salesforce Account

Create and activate a Salesforce Developer Org using the required developer signup and activation process.

Milestone 2: Data Management – Custom Object

Object Name: Support Ticket Intelligence

API Name: Support_Ticket_Intelligence__c

Field Label	API Name	Data Type	Description
Ticket Number	Ticket_Number__c	Auto Number	TKT-{0000}
Customer	Customer__c	Lookup (Account)	Related Account
Contact	Contact__c	Lookup (Contact)	Customer contact
Issue Type	Issue_Type__c	Picklist	Technical, Billing, General
Description	Description__c	Long Text Area	Issue details
Priority Level	Priority_Level__c	Picklist	Low, Medium, High
Status	Status__c	Picklist	New, In Progress, Resolved
Created Date	Created_Date__c	Date	Ticket date
Assigned To	Assigned_To__c	Lookup (User)	Support agent
SLA Breach Risk	SLA_Breach_Risk__c	Checkbox	Risk flag
Resolution Time (hrs)	Resolution_Time__c	Number	Time taken

6.1 Auto-Launched Flow

Flow Type: Auto-Launched Flow (No Trigger or called from Agentforce).

Variable	API Name	Type	I/O
Account Name	varAccountName	Text	Input
Account Id	varAccountId	Text	Output
Ticket Id	varTicketId	Text	Output
Priority Level	varPriorityLevel	Text	Output
Assigned Agent	varAssignedTo	Text	Output
Action Message	varActionMessage	Text	Output

6.2 Flow Implementation

Step	Element	Configuration
1	Get Account	Account; Name = varAccountName; CreatedDate DESC; first record
2	Store Account Id	varAccountId = Get_Account.Id
3	Get Ticket	Support_Ticket_Intelligence__c; Customer__c = varAccountId; CreatedDate DESC; first record
4	Store Ticket Id	varTicketId = Get_Ticket.Id
5	Analyze Description	High: urgent/not working/failure; Medium: issue/slow/delay; default: Low
6	Set Priority	Set varPriorityLevel to High, Medium, or Low
7	High Priority Check	varPriorityLevel = High
8	Create Task	Subject Urgent Ticket Handling; WhatId Ticket Id; Priority High; Status Not Started

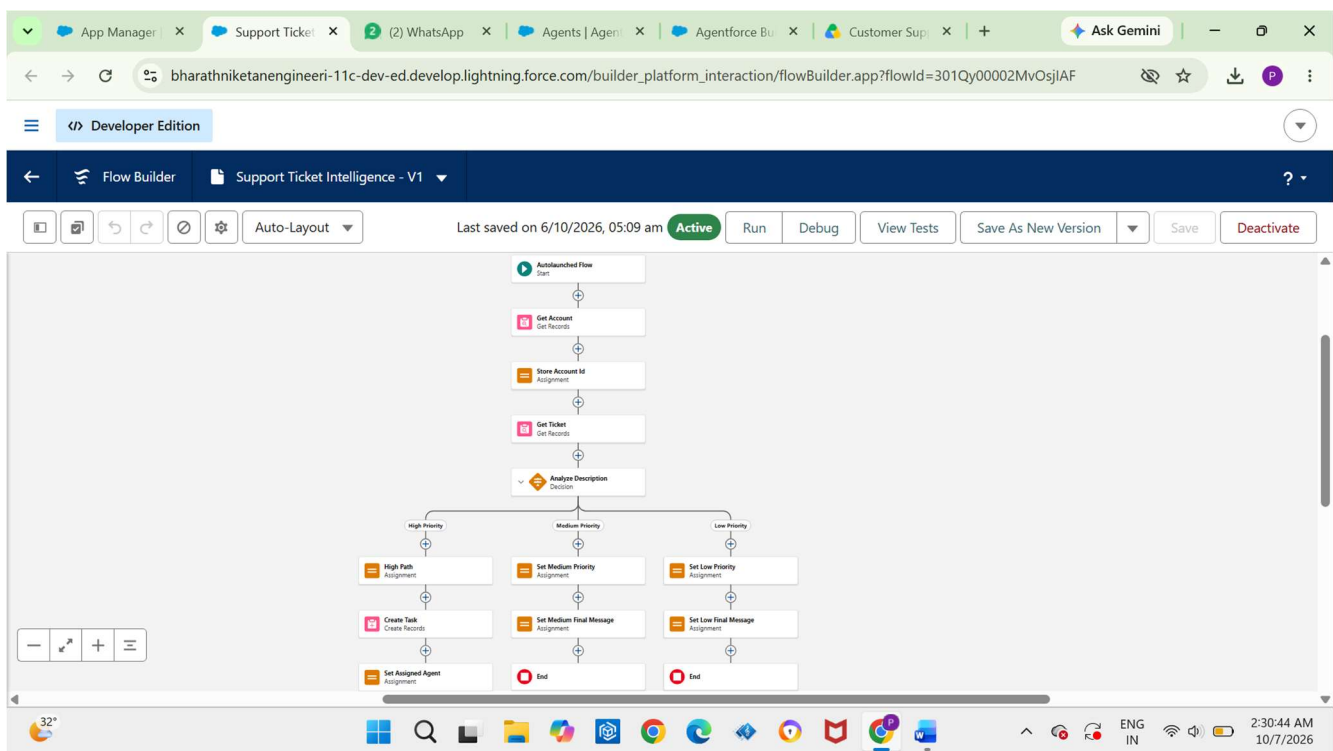
9	Assign Agent	varAssignedTo = Senior Support Agent
10	Optional SLA Check	Created Date older than 2 days → optional internal flag/message
11	Final Message	High: urgent handling; Medium: handled shortly; Low: queued
12	Save Flow	Support Ticket Intelligence

6.3 Final Flow Structure

Start → Get Account → Assignment (Account Id) → Get Ticket → Assignment (Ticket Id) → Decision (Analyze Description) → Assignment (Priority Level) → Decision (High Priority) → Create Task → Assignment (Assigned Agent) → Optional SLA Decision → Assignment (Final Message) → End

6.4 Final Flow Outcome

- Takes Account Name as input.
- Fetches latest support ticket.
- Analyzes description using keywords.
- Sets priority automatically.
- Creates task for urgent cases.
- Returns output to Agentforce.



7. AGENTFORCE CONFIGURATION

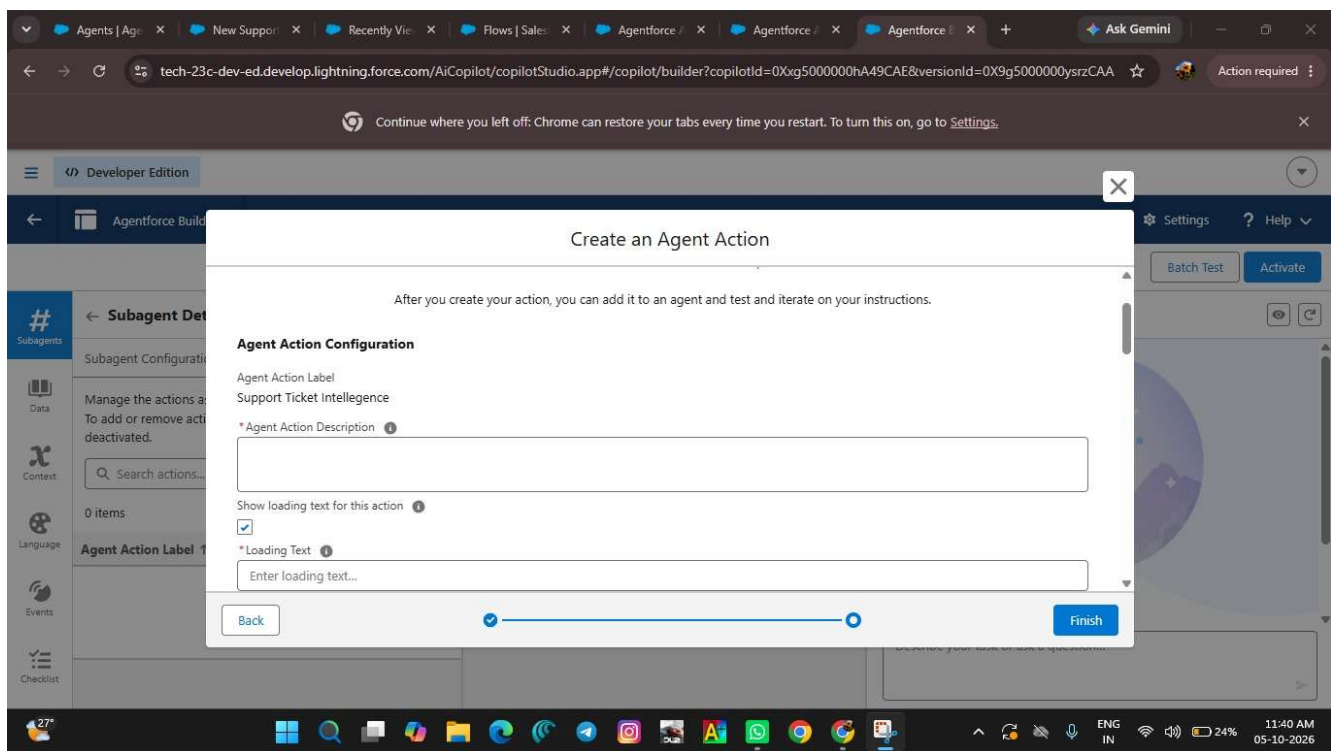
7.1 Agentforce Subagent

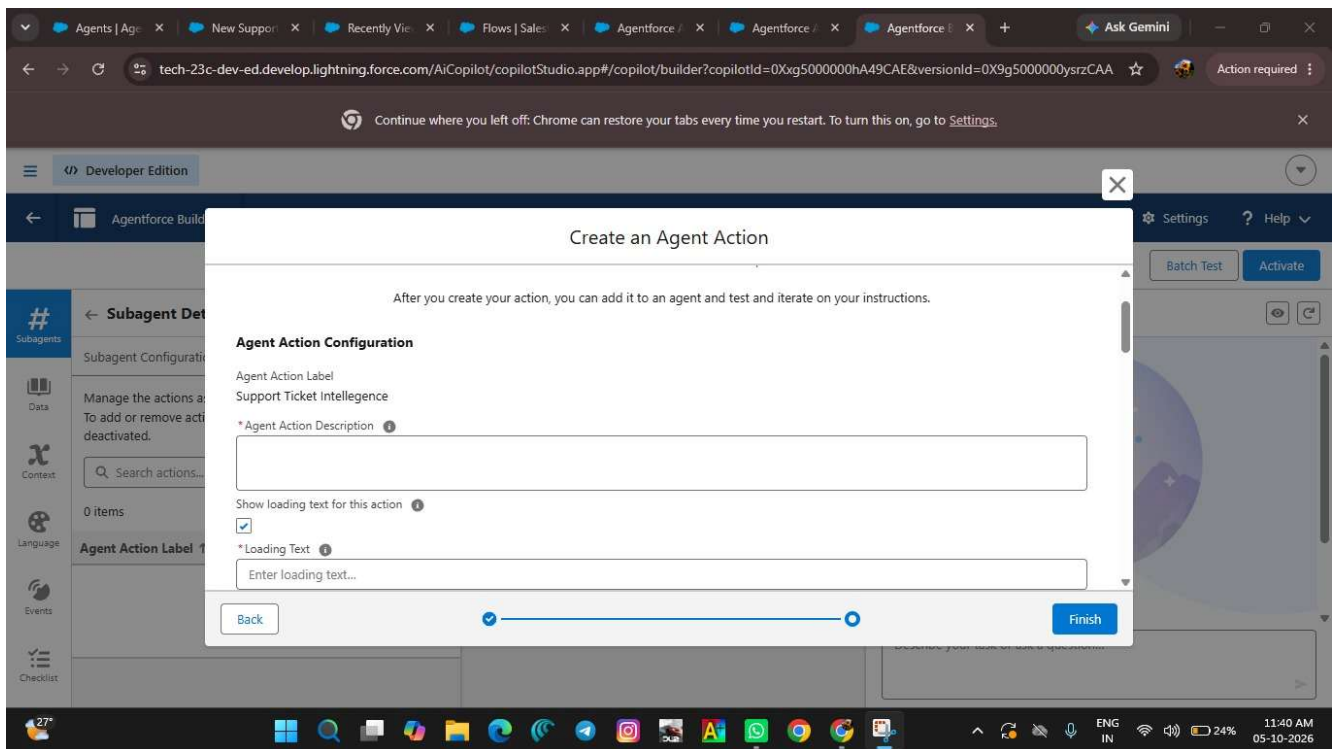
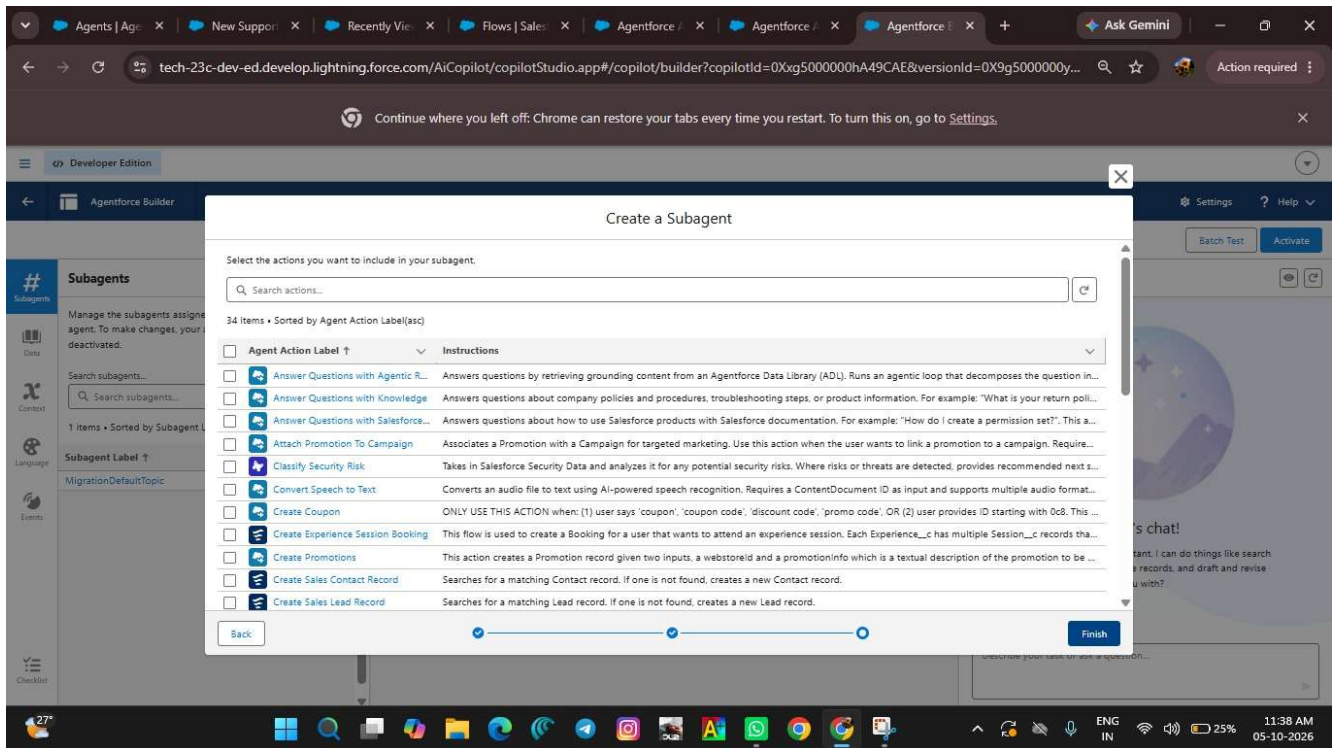
Configuration	Value
Name	Support Ticket Priority Analysis

API Name	Support Ticket Priority Analysis
Classification Description	Analyzes customer support tickets to determine priority based on issue description and urgency.
Scope	Only analyze ticket descriptions, determine High/Medium/Low, and trigger backend automation for task assignment; unrelated requests are outside scope.

7.2 Instructions

- Ask for Account Name if not given.
- Retrieve the latest support ticket associated with the Account.
- Read and analyze the ticket description.
- High: urgent, not working, failure.
- Medium: issue, slow, delay.
- Low: none of the above.
- Set priority accordingly.
- For High, trigger the Auto-Launched Flow and ensure a task is created.
- Assign appropriate support level, e.g. senior agent for High.
- Return clear High/Medium/Low action messages.
- Do not update records directly or send emails; use Flow for actions.





7.3 Agent Action

Description: Analyzes customer support tickets based on Account Name, determines priority using issue description, and triggers automation for high-priority cases.

Type	API Name	Data Type	Setting
Input	varAccountName	lightning__textType	Require Input; Collect Data From User
Output	varAccountId	lightning__textType	Show in Conversation
Output	varActionMessage	lightning__textType	Show in Conversation
Output	varAssignedTo	lightning__textType	Show in Conversation
Output	varPriorityLevel	lightning__textType	Show in Conversation
Output	varTicketId	lightning__textType	Show in Conversation

8. FUNCTIONAL & PERFORMANCE TESTING

Test Case	Condition	Expected Result
TC-01	Description contains urgent	High; urgent task; senior support handling
TC-02	Description contains not working	High; urgent task
TC-03	Description contains failure	High; urgent task
TC-04	Description contains issue	Medium; handled shortly
TC-05	Description contains slow	Medium; handled shortly
TC-06	Description contains delay	Medium; handled shortly
TC-07	No configured keyword	Low; queued
TC-08	Older unresolved ticket, SLA enabled	SLA risk flag/message
TC-09	Agentforce receives Account Name	Latest ticket and priority/action outputs returned

9. ADVANTAGES & DISADVANTAGES

Advantages

- Automatic Ticket Prioritization
- Faster identification of critical issues
- Reduced Manual Effort
- Intelligent Ticket Handling
- Automatic Urgent Task Creation
- Improved Support Assignment
- Conversational Support
- Better Response Time
- SLA Risk Awareness
- Scalable Support Operations
- Improved Team Productivity

Disadvantages

- Keyword-Based Priority Logic
- Limited Understanding of Context
- Dependence on Accurate Ticket Data
- Flow Configuration Dependency

- Agentforce Configuration Dependency
- Limited Scope of Agentforce
- SLA Monitoring is Limited
- Assignment Logic May Need Expansion
- Maintenance Required
- Limited Analytics in Current Scope

10. CONCLUSION

The Customer Support Ticket Priority Prediction and Automated Assignment System demonstrates how Salesforce Flow and Agentforce can work together to reduce manual support-ticket handling. The solution retrieves the latest ticket for an Account, analyzes its description, classifies priority, creates an urgent task for High-priority cases, assigns a support level, and returns a clear conversational response. This supports faster ticket handling, reduced manual effort, intelligent prioritization, and better team productivity.

11. FUTURE SCOPE

- Add more priority indicators and business rules.
- Extend Agentforce outputs with additional ticket information.
- Add more sophisticated SLA monitoring and escalation.
- Support additional support teams or assignment rules.
- Introduce analytics for priority trends and resolution performance.
- Add more variables and configure them in the Agentforce action for richer outputs.

12. APPENDIX

- A. Custom Object: Support Ticket Intelligence
- B. Key Salesforce Records: Account, Contact, User, Task
- C. Flow: Support_Ticket_Intelligence – Auto-Launched Flow
- D. Agentforce Subagent: Support Ticket Priority Analysis
- E. Agentforce Action: Flow-based Support Ticket Intelligence action